Tian-Yu Ma

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Research interests: Brain-computer interfaces, neural decoding, AI for Neuroscience / Scientific ML

Education

University of Chinese Academy of Sciences | PhD student, Neuroscience 2022 - 2027

Doctoral research at CEBSIT, CAS; expected graduation in summer 2027.

School of Life Sciences, Nanjing University | Undergraduate 2018 - 2022

Chemistry and Life Sciences (2018-2019); Life Sciences (2019-2022). GPA: 4.28/5.00 (17/46); junior year: 4.45/5.00 (8/46).

Selected courses (out of 100): Neurobiology 87.5, Physiology 93, Microbiology 97, Biostatistics 95, Biochemistry 90, Molecular Biology 90.

Research Experience

Doctoral research | Closed-loop BCI learning 2022 - present

Center for Excellence in Brain Science and Intelligence Technology, CAS

• Investigate neural activity during closed-loop BCI learning, combining multi-region electrophysiology and behavioral analysis to study how the brain learns to control external systems.

• Explore whether microcurrent stimulation modulates learning speed in a mouse BCI task.

• Use Python for neural signal processing, data analysis and statistical modeling.

• Doctoral research manuscript in preparation; poster accepted for SfN 2026.

Neural Interface Group | Zhengtuo Zhao's lab 2021 - 2022

CEBSIT, CAS | Undergraduate thesis and research internship

• Contributed to nonhuman-primate BCI experiments and data analysis; co-author of an Advanced Science paper (2023).

Nanjing University | Jing-Ning Zhu's lab 2019

• Participated in neuroscience and animal behavior research; co-author of a PNAS paper (2020).

Publications

1. Tian, Y., et al. (including Ma, T.). (2023). An Ultraflexible Electrode Array for Large-Scale Chronic Recording in the Nonhuman Primate Brain. Advanced Science, 10(33), 2302333. doi:10.1002/advs.202302333.

Co-author (8th author); contributions include animal behavioral experiments, neural recording and data analysis.

2. Zhang, X.-Y., Peng, S.-Y., Shen, L.-P., Zhuang, Q.-X., Li, B., Xie, S.-T., Li, Q.-X., Shi, M.-R., Ma, T.-Y., Zhang, Q., Wang, J.-J., & Zhu, J.-N. (2020). Targeting presynaptic H3 heteroreceptor in nucleus accumbens to improve anxiety and obsessive-compulsive-like behaviors. PNAS, 117(50), 32155-32164. doi:10.1073/pnas.2008456117.

Co-author (9th author); neuroscience research conducted during undergraduate studies at Nanjing University.

Projects and Leadership

NJU-China | iGEM team leader 2020 - 2021

• Led project and experimental design for an exosome-mediated siRNA delivery strategy targeting lung cancer; coordinated team experiments and scientific communication.

• Team received the iGEM 2020 Silver Medal and CCiC Best Poster Award. Project archive.

Internet+ Innovation and Entrepreneurship Competition 2021

• Lead member of the project on in vivo self-assembly of targeted siRNA drugs; national bronze award.

National University Student Innovation Program 2019

• Lead member of a research project on the effects of oxytocin on the cerebellum and autism.

Technical Skills

• Experimental neuroscience and BCI: closed-loop behavioral paradigms, multi-region electrophysiology, neural recording, brain-control decoding and experimental design and animal behavioral experiments.

• Data and modeling: Python, neural signal processing, time series analysis and statistical modeling; neuronal tuning, population trajectories, LFP power/coherence, GLM, RRR and CCA.

• AI-assisted development / Vibe Coding: prototype research tools and software with AI assistance, checking outputs against experimental requirements and data; continuing to develop machine learning and scientific computing skills.

Honors and Awards

• UCAS Merit Student: academic years 2023-2024 and 2025-2026.

• iGEM 2020 Silver Medal; CCiC 2020 Best Poster Award; Internet+ 2021 national bronze (team awards).

• Nanjing University: Second-Class People’s Scholarship; Second Prize, 23rd Forum of Sciences & Art; Third Prize, Academic Salon for Top Innovative Talents.